

NDNSF: A Data-Centric Service Framework for Secure, Collaborative, and Dynamic Networked Applications

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Why Service-Level Mechanisms Are Needed

Networked applications need mechanisms for service discovery, invocation authorization, and application-data exchange.

- ▶ Some operations discover a currently willing Provider; others must retain an already selected Provider.
- ▶ Applications must associate permissions, execution state and exchanged data with the requested operation.
- ▶ Changes in service availability, resources and connectivity add further challenges.

Research goal: a reusable service protocol. UAV and inference applications provide validation cases, not limits on the framework's intended scope.

Research Gap and Two Research Scopes

Foundation: evaluate discovery and authorization

The initial prototype discovers candidates and selects one Provider whose positive ACK passes validation. Discovery confidentiality, service permissions and management costs matter even without collaboration.

Extension: coordinate complementary roles

Some tasks need several Providers with different responsibilities. Separate calls do not define a shared role assignment or enforce dependencies between them.

Research gap: existing invocation mechanisms do not by themselves bind runtime Provider assignments to named-Data dependencies and recovery. Plan-commit support is partial [28]; the proposed research completes these rules and tests their reuse and cost.

NDN Primitives and Service-Level Decisions

NDN building block	Remaining application responsibility
Named Data and signatures [1]	Apply Trust Schema [16] rules to validate signing authority
Interest retrieval and caching [1,26]	Check task identity, dependencies and deadlines; a Content Store is not persistent application storage
Dataset synchronization / SVS [2,3]	Synchronize names and state; the application defines service-Data transitions
NAC [4] / enhanced NAC-ABE [19]	Encrypt Content for read access; check invocation permission separately

NDN: Named Data Networking [1]; NAC: Name-based Access Control; ABE: attribute-based encryption; SVS: State Vector Sync.

NDNSF adds service authorization and invocation-state rules.

Research Scope: From Application Needs to Checks

Follow the invocation lifecycle, informed by invocation [9,10] and authorization [12] mechanisms.

Lifecycle stage	What can fail	Focus
Discover candidates	Ineligible candidate or unauthorized descriptor access	Authorization
Select execution	Wrong Provider, unauthorized or repeated start	Authorization
Use inputs	Wrong producer, object or dependency	Dependency
Close or replace	Late results advance the wrong execution	Recovery
Change permissions	Missed grant; withdrawn permission still accepted	Permissions

These lifecycle checks motivate the three research questions introduced on the next slide.

Three Research Questions

- ▶ **RQ1:** When does ABE-backed authorization offer a favorable confidentiality–management-cost tradeoff over a role-constrained alternative inspired by DNMP [12] for protected Provider discovery and invocation, as permission scale, overlap, membership changes and withdrawals vary?
- ▶ **RQ2:** After ACK collection closes, how can single-Provider invocation be extended so runtime-selected Providers receive explicit roles and named dependency scopes, and accept producer- or cache-retrieved Data only when it matches the committed execution context?
- ▶ **RQ3:** Under what workload, network and failure conditions do ABE-backed discovery, role/dependency admission and bounded recovery improve completion and latency, and when do their costs outweigh the benefits?

RQ1 studies invocation; RQ2 studies collaboration. RQ3 evaluates their benefits and costs, not another protocol.

Related Work: Mechanisms and Comparison Boundary

Prior work	What this study compares
NSC [10]: signed invocation	Caller authorization versus Provider predecessor checks
DNMP [12]: role commands	Standing permission versus Provider task assignment
CFN [33]: dynamic graph	Trusted components versus Provider admission after role/edge checks
CaSCON [15]: composition	Cache-assisted composition versus Provider admission and replacement recovery
NFN [6], NFaaS [7] and RICE [9] are computation/invocation precedents.	

Focus: downstream Providers check role, predecessor, edge and execution state before accepting Data; replacement reuses only permitted artifacts.

Research Hypothesis and Expected Contributions

Hypothesis: A common named-Data protocol can separate discovery from authorized execution and enforce runtime role assignments when intermediate objects are used.

Intended knowledge	Evidence to deliver
When permission reuse is worthwhile	Permission overlap, churn and lifecycle costs under equal protection
Which task-specific checks are needed	Valid completion; wrong-role/edge and replacement tests
Which rules can be shared	Common NDN-SF-runtime checks; application validators; matched workload tests

Contribution: evaluated service-level mechanisms and evidence, not a new encryption primitive or general workflow optimizer.

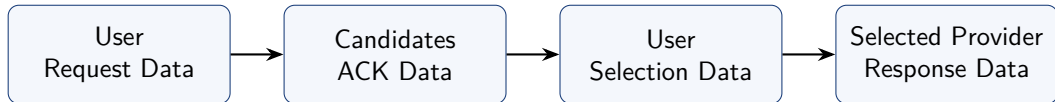
Protocol Roles and Trust Bootstrap

- ▶ User: requests a service and selects a Provider; the extension also assigns collaboration roles.
- ▶ Provider: offers and executes a service. A node may play both roles.
- ▶ Controller: administers identities and service permissions approved by the organization.
- ▶ Participants use provisioned trust anchors and validation rules for certificates and Controller-signed authorization data.

Enrollment establishes identities and service permissions. The User then discovers which authorized Providers are currently willing and selects participants for the task.

Example Invocation: Request a Timely Aerial Image

An operator requests an aerial image. Authorized camera Providers A and B consider the task.



- ▶ A returns a positive ACK; B is busy and returns a negative ACK.
- ▶ The User selects A; A checks Selection before capturing the image.

Arrows show application order. Data is retrieved with Interests; neither an ACK nor a network delivery alone authorizes execution.

Single-Provider Invocation: Four NDNSF Data Objects

Data type	Purpose in this workflow
Request Data	Describe required service and participation constraints; include protected discovery information and RequestNonce
ACK Data	Report positive or negative participation; positive ACK supplies selection information and ProviderChallenge
Selection Data	Assign selected execution; provide input/references and protected request-scoped keys; constructed using ProviderChallenge
Response Data	Return status, results or result references; the User still validates them before use

Sync helps discover names; Interests retrieve Data. The Provider checks Selection before execution; a positive ACK alone does not start the task.

Discovery Descriptor versus Post-Selection Input

Discovery descriptor

Needed to decide whether to participate: area, deadline, sensor/model and resource requirements.

Application input

Needed to execute the selected task: images, detailed processing parameters, or protected input references. Input and results use request-scoped symmetric keys. Selection delivers key material encrypted for the selected Provider; the requesting User retains the keys.

The boundary follows necessity. A location needed to assess coverage belongs in the protected descriptor; an image needed only for processing can wait.

Three Security Requirements; One Additional Benefit

1. Keep discovery descriptors confidential from unauthorized readers before selecting a Provider.
2. Authenticate participants, verify service permissions, and reject replayed invocation messages.
3. Grant and withdraw service permissions during operation.

Additional benefit: aggregate service permissions in a decryption key (DKEY) while retaining identity-signing credentials. This is a management property, not a separate security claim.

Reason 1: Confidential Discovery Before Selection

- ▶ The User knows the service, but not yet the currently willing Provider identities.
- ▶ An observation area and deadline can reveal intended activity even without the full application input.
- ▶ The User encrypts descriptor Content under the service-authorized ABE policy, without listing current Provider identities. [19]

Confidential from parties lacking the required decryption capability. Authorized but unselected Providers can read it; names and traffic patterns remain visible.

Reason 2: Reuse Attribute-Based Service Permissions

- ▶ NDNFSF uses ABE to keep discovery descriptors non-public before selection. [19]
- ▶ NDNFSF reuses the attribute authority to manage Provider and User service permissions. [28]
- ▶ RequestNonce and ProviderChallenge checks use these permissions during invocation.

ABE discovery plus role-based invocation is also possible. NDNFSF avoids adding role signing credentials solely for service membership; signature validation and request-state checks remain necessary. User DKEYs and challenge processing still add cost.

How NDNSF Checks Service Permissions

1. Provider permission: recover RequestNonce from ABE-protected Request content and return it in ACK Data. The User compares it with RequestNonce stored for this request.
2. User permission: recover ProviderChallenge from ABE-protected ACK content and construct Selection Data. The Provider validates Selection using its stored ProviderChallenge.

Participants also check current service permissions. Challenge checks establish access to protected values, not exclusive key possession. RequestNonce is an application value, distinct from the Interest Nonce.

Identity and Replay Checks Have Different Jobs

Identity authentication

Validate Data signatures and certificate chains under trust rules. An accepted signing identity alone does not establish service permission.

Replay protection

Match messages to an active request and its state. Record whether ProviderChallenge has already been used to accept Selection for execution; duplicate Selection must not start another execution.

Retain or safely invalidate request state, including across restart. Encryption or a challenge value alone does not prevent replay.

DNMP: Trust-Schema Command Authorization

Distributed Network Measurement Protocol (DNMP) [12] authorizes measurement commands:

1. The Network Key anchors role keys, device keys, and the schema.
2. The client uses target, probe type, arguments, and user role to find an allowed signing key; among matching keys, it chooses the least privileged.
3. The receiving Network Observer Daemon validates the publication under the schema before dispatching the probe.

DNMP also uses publication lifetimes and retained state against replay, assuming loosely synchronized clocks. These are concrete command-authorization mechanisms, not just identity checks.

DNMP-Inspired Role Authorization versus NDNFSF

Design	How execution is authorized	Discovery protection
DNMP-inspired [12]	Role-constrained signer plus request state	Same ABE descriptor (controlled comparison)
NDNFSF ABE-backed	Signature, challenges, permission and request-state checks	Service-level ABE

Both designs retain the same ABE-protected discovery, Selection, current-permission and replay checks. This adapts DNMP's role-constrained signing idea; it does not reproduce DNMP. Group encryption is separate.

RQ1: Evaluate Authorization Design Tradeoffs

Model: one Controller domain, one ABE generation; n Users, k services/User, and r role certificates/User. Additional objects only.

Authorization arrangement	Additional objects for n Users
Per-service roles ($r = k$)	$nr = nk$: $O(nk)$ certificates
One composite role per User	$nr = n$: $O(n)$ certificates
One policy DKEY per User	n : $O(n)$ DKEYs; each lists k services

For fixed k , $O(nk) = O(n)$. 500 versus 100 is inventory only—not an asymptotic or total-cost advantage.

Separate policy size, cryptographic work, bytes, setup, invocation, grants, withdrawals and maintenance.

RQ1: Three Planned Validation Configurations

Scenario	Example	Compare
G1: Per-service roles	100 Users; four disjoint five-service bundles	500 role certificates versus 100 User DKEYs
G2: Combination roles	Same matrix; one bundle-role certificate per User	100 role certificates versus 100 DKEYs
G3: Withdrawals	Repeat G1/G2; fixed call rate, varied withdrawal rate	Rekey traffic, update delay and completion

These are analytical examples, not measurements. Both retain the same discovery protection, permissions and status-validity bound; User-call and Provider-discovery withdrawal are separate cases.

Identify when each design is preferable, with correctness and availability checked alongside cost.

Runtime Permission Grants and Withdrawals

- ▶ Both operations update Controller-signed permission status. [28]
- ▶ Ordinary grant: update the target's complete policy; it retrieves a replacement DKEY. Public parameters stay unchanged.
- ▶ Withdrawal: rotate the Controller's ABE master key and public parameters; remaining members obtain DKEYs for their remaining permissions.

The master key stays with the Controller. Rotation is Controller-wide. Participants must retrieve and validate updates before applying them; disconnected nodes cannot do so immediately.

Proposed Permission-Update Path: Discovery, Validation and Application

1. Revised design: the Controller publishes new public parameters as signed Data; Sync announces availability.
2. Participants retrieve and validate parameters and permission status, then refresh DKEYs as required.
3. Grants also require status discovery, even when public parameters do not change.

Status retrieval and rotation are implemented; Sync announcements are planned for end-to-end validation. Evaluation will measure propagation, rejection delay and rekeying traffic. Previously disclosed keys and plaintext remain accessible to their holders.

Threat Model: What Is Trusted?

- ▶ Trusted: Controller authorization and the User's planning decision.
- ▶ Untrusted delivery: an object may arrive through a producer, forwarder, cache, or repository and still requires validation.
- ▶ Adversarial cases to test: wrong identities, permissions, challenges, request state, roles, dependencies, or execution identities.
- ▶ Not established by signatures/ABE: honest resource reports, correct inference, truthful images, or resistance to authorized key sharing.

Scope: trusted Controller and User, Content confidentiality, and withdrawal after validated updates. Malicious-coordinator agreement and traffic-analysis resistance are outside this scope.

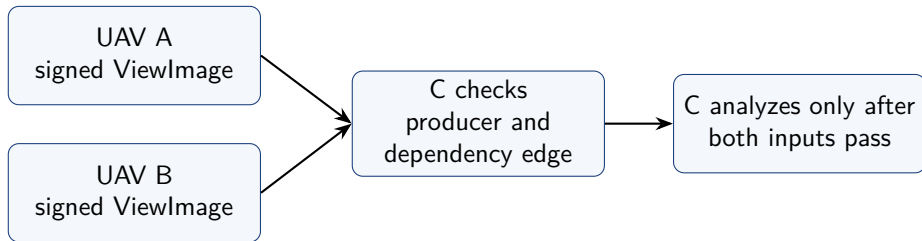
RQ2 Extension: From Positive ACKs to a Role Plan

1. Application supplies roles, dependencies, constraints and deadline.
2. The User closes ACK collection and freezes eligible, validated positive ACKs.
3. A replaceable planning strategy assigns Providers; the User signs the resulting plan.
4. Providers validate assignments, permitted inputs and resource readiness.

The role template is defined before discovery; concrete signer and name constraints are filled from validated ACKs. Current plan-commit checks are partial [28]; dependency and recovery contracts remain to be validated.

Plan Binding: Roles and Named Dependencies

Proposed task: User assigns UAV A to view 1, UAV B to view 2, and Provider C to analysis.



Arrows show object dependencies; Interests retrieve the images from producers or caches.

A valid signature and matching request ID do not let B's image fill A's view-1 role. C must check the committed assignment and declared edge before accepting it.

Proposed Plan-to-Data Admission Checks

1. Validate the Data signature using the appropriate trust rules.
2. Check active execution and deadline; identify the permitted dependency.
3. For new Data, match request/plan/execution and assigned producer; for historical Data, check explicit reuse permission.
4. Check that the object matches the declared edge, permitted name, data format and key scope.
5. Require complete validated input before the dependent computation starts.

The proposed rules apply equally to Data retrieved from producers, caches and repositories. End-to-end tests will check valid inputs and rejection of wrong-role, wrong-edge and outdated-execution objects.

Collaboration Recovery Needs Explicit State

Event	Required response
Late ACK	Do not silently change the committed plan
Duplicate Selection	Check prior authorization; do not repeat side effects
Incomplete dependency	Wait, recover, or fail within deadline; partial data is not sufficient
Provider replacement	Authorize new assignment; reject old actions; define permitted reuse
Cancellation	Reject late messages after cancellation

A replacement plan may reuse one old artifact under its original Name and signature; it does not reactivate the old execution. Application checks still decide usability.

NDNSF-UAV: Planned Framework Validation

UAV (unmanned aerial vehicle) workflows test NDNSF mechanisms. [28]

Application function	NDNSF checks under study
Control, telemetry, mission status	Provider/request association, permissions, operation lifecycle
Live video and recordings	Stream lifecycle, protected delivery, named-object retrieval
Detection and multi-view tasks	Service provision/consumption, role assignments, dependencies

Use these workflows to test the invocation foundation and collaboration extension. Validate service behavior through the actual NDNSF path; evaluate flight safety and perception accuracy separately.

UAV Collaboration Example: Capture, Analyze and Return

An operator selects a ground area on a map and requests views and analysis.

1. Selected UAVs capture views, symmetrically encrypt images, and publish signed ViewImage Data under their own authorized namespaces.
2. The assigned compute Provider retrieves permitted images and publishes Annotation Data.
3. The compute Provider returns encrypted Annotation Data with original image Names and, when authorized, key material for the requester.

Reuse original image Data. Fixed-camera viewing geometry requires synchronized UAV attitude and calibration; GPS alone does not establish visibility.

UAV Authorization and Safety: Separate Responsibilities

- ▶ NDNF runtime: service authorization and request state.
- ▶ UAV application: operator authority, mission progress and reassignment.
- ▶ Vehicle: readiness, flight limits and local failsafe behavior.

The ground station currently checks operator authority leases; the Drone MAVLink handler does not independently repeat that check. [28] Tests will examine this command path. NDNF ACK Data reports willingness; flight-controller acceptance and vehicle behavior are separate observations.

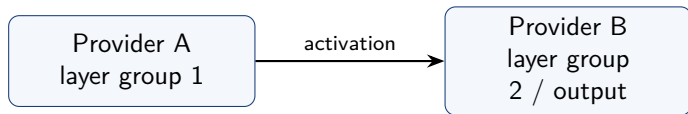
NDNSF-DI: Planned Framework Validation

Existing native lifecycle: prepare a model, submit an inference request, and collect results. [28]

Application function	NDNSF checks under study
Single-Provider inference	Discovery, permissions, input and result delivery
Model preparation and reuse	Package identity, artifact references, named-object retrieval
Incremental output and conversation	Request state, cancellation, output lifecycle
Multi-Provider inference	Assignments, dependencies, recovery

PreparedModel describes a verified local package. Provider selection and remote preparation follow during invocation. The study will extend focused tests to the selected model, device and recovery configurations.

DI Collaboration Example: Named Computation Dependencies



- ▶ Verify a local model package; choose partitions and Providers after ACK collection closes.
- ▶ Providers authenticate Selection before preparing or reusing assigned resources.
- ▶ Pipeline divides layer groups; tensor partitioning needs supported collective exchanges.
- ▶ ONNX [25] describes the graph; Repo [24] stores artifacts. Neither implements distributed execution by itself.

Minimum planned case: a two-Provider partition; larger pipelines are extensions. The contribution is checked runtime assignment and named-data exchange, not model partitioning itself.

DI State and Planned Validation

- ▶ Model weights, reusable artifacts, request activations and key/value (K/V) state have different lifetimes. [31]
- ▶ With K/V caching, the runtime reuses prior tensors and supplies next-step inputs; resident weights do not replace this state.
- ▶ Cross-process tests will compare distributed results with the same model's unsplit execution.
- ▶ Begin with one supported model/partition; evaluate additional strategies and LLM streaming as their implementations become ready.

Earlier small-model measurements motivate separate preparation, transfer and execution measurements in the new campaigns. [28]

Cross-Application Study: Framework and Application Responsibilities

Framework checks	Application-specific checks
Signer, assigned role and edge	UAV: view/time requirements; DI: model/tensor compatibility
Complete named object; permitted reuse	Decide whether the image or retained state remains useful
Cancellation/replacement state	Stop physical work or drain computation and release resources

Test both applications with wrong-role inputs and authentic but application-invalid inputs. Record which layer rejects them; retain valid controls.

Expected outcome: shared checks and explicit application validators. Signatures establish provenance, not image truth or numerical correctness.

Validation Stages: What Each Test Establishes

Test stage	What it establishes
Contract / local tests	Valid transitions and explicit rejection cases
MiniNDN [22]	Authenticated request chains, caches, loss and outages
UAV simulation / bench	Operator workflow, telemetry, camera and command handling; PX4 SITL [23] is simulation
TigerCluster [28] (preparation)	Multi-node GPU inference: correctness, latency, data-transfer overhead and peak GPU memory
Supported devices	Actual camera/inference outputs, memory and processing costs

Use defined outcomes and reproducible configurations. Cluster experiments complement UAV wireless and mobility tests; simulation does not establish real-flight safety.

RQ3: Define the Baselines Before Comparing

- ▶ NSC [10]: named service calls; describe the tested executor, timeout, queue and recovery behavior.
- ▶ gRPC [17]: describe resolver, sequential/parallel policy, deadlines, retry and cancellation.
- ▶ Match Provider count, capacity, task, topology/traces, publication times and measurement windows.
- ▶ Report fixed-list and dynamically discovered configurations as different configurations.
- ▶ Specify primary outcomes, completion deadlines and useful-improvement criteria before comparative runs.

These observations characterize the existing invocation foundation; they do not answer collaboration or recovery. Comparative conclusions will refer to the tested configurations.

Preliminary Evidence: Completion by AP Radius

Ten seeds (62–71), four Providers/system, 2 m/s; mean seed completion rate. [28]

AP radius	NDNSF	gRPC	NSC
50 m	54.57%	53.80%	55.47%
100 m	97.97%	97.97%	98.33%
150 m	100.00%	100.00%	100.00%

Observed completion rates are close across systems. The next slide examines switching latency in a separate holdout campaign (seeds 72–81).

Preliminary Evidence: Switching Latency

Seeds 72–81; four Providers/system; 100 m, 2 m/s; 1,312 aligned switching requests. [28]

Comparison	Seed-p95 difference	95% CI
NDNSF minus gRPC-SEQ-4	−597 ms	[−992, −200] ms
NDNSF minus NSC-4	−2618 ms	[−2916, −2315] ms

Completed: NDNSF 1,311; gRPC 1,297; NSC 1,296 (of 1,312 each).

NDNSF has lower mean seed-p95 latency in this switching workload. Each p95 uses that system's successful requests; different failure sets limit interpretation.

Preliminary Evidence: Execution Starts

Six-seed, capacity-matched comparison; 1,800 requests per system. [28]

Configuration	Completed requests	Execution starts
NDNSF	1,798 / 1,800	1,800
gRPC-PAR-4	1,800 / 1,800	7,180

NSC-4 also completed 1,800/1,800, with 1.002 starts/request (rounded). No work-efficiency advantage over NSC is claimed.

NDNSF used fewer execution starts than gRPC-PAR-4 at the reported completion rates. Bytes, CPU time and energy require separate measurement.

Proposed Evaluation: Tests for the Research Claims

RQ	Test setup	Observable decision
RQ1	1 User, 4 candidates; same discovery encryption	Reject unauthorized starts; compare authorization costs
RQ2	Wrong role/edge; direct/cache; remove-one-check	Reject invalid dependency before work
RQ2/3	Cancel/replace one Provider	Reject late old work; measure recovery
RQ3	Fixed inputs; UAV relay; feasible unsplit DI	Compare outputs, completion, latency, bytes and memory

Retain signatures and request IDs. Define tolerances and improvement criteria before runs.

Remaining Work and Scope Controls

Minimum case	Evidence required
UAV	Two capture roles and one analyzer; fixed images/algorithm; compare direct exchange with User relay
DI	Target two-Provider YOLO partition; declare crossing tensors; fix model/cut after compatibility checks
Both graphs	Valid completion, wrong-input rejection, one cancellation/replacement case, matched costs
Extensions	More models, LLM streaming, larger deployments and real-flight trials

These are planned cases. If either core graph cannot be validated, reconsider its claim and schedule with the committee; optional features cannot replace missing core evidence.

Completion Plan: Spring 2027 Defense Target

Target date	Deliverable
September 2026	Agree scope/RQs; freeze one UAV graph and one DI partition; resolve correctness blockers
October–November 2026	Complete registered experiments; freeze evidence and write bounded RQ answers
December 2026	Complete integrated advisor-review draft
January 2027	Committee review and readiness decision; reduce scope or revise timing if core evidence is missing
February 2027	Final dissertation defense, subject to committee approval

Committee draft: two weeks before defense. Spring 2027 application/candidacy: February 5; final ProQuest: March 26. Recheck before filing. [29,30]

Expected Research Outcomes

- ▶ **RQ1 answer:** tradeoff regions across permission overlap, membership changes and withdrawals, using complete alternatives with equal protection.
- ▶ **RQ2 answer:** which assignment and dependency rules can be shared across UAV and DI, what they reject, and their recovery limits.
- ▶ **RQ3 answer:** measured benefits, negative results and deployment limits, including which checks belong in the NDN SF runtime or the application.
- ▶ Reproducible configurations linking each conclusion to tested scope.

The intended outcome is a tested service protocol and an evidence-backed account of when its mechanisms are worth their cost.

Discussion: Is This Scope Sufficient and Feasible?

1. Are the three research questions specific enough to support defensible answers?
2. Are the authorization alternatives complete and fairly configured?
3. Are one UAV graph and one supported DI partition sufficient to test the abstraction?
4. Which optional features should be excluded if they delay the core evidence?

Committee feedback sought: research significance, bounded scope, evaluation criteria, and a feasible completion plan.

Backup: Conditional Permission Aggregation

- ▶ Within one Controller domain, combine several service permissions in one entity's DKEY; retain identity-signing credentials. [19,28]
- ▶ DNMP represents command authority through role keys and schema rules. One role can permit several commands. [12]
- ▶ Where equivalent permissions require distinct role credentials, NDNFSF avoids maintaining those extra signing credentials.

Fewer independently managed credentials does not imply fixed DKEY size or lower total cost. Policy size, key distribution and updates still matter.

Backup: When Alternatives May Be Preferable

- ▶ Known Providers and simple rules: signed invocation plus recipient encryption may be sufficient.
- ▶ A small, stable receiver group: ordinary group-key management may be attractive.
- ▶ Rich command-argument restrictions: DNMP-style rules make those constraints explicit. [12]
- ▶ Many changing entitlements: measure both policy/key updates and credential/rule updates before claiming a management advantage.

Compare complete combinations of authorization and encryption. ABE-backed authorization is one design choice, not a universally superior solution.

Backup: Archived Small-Model Diagnostic

Earlier prototype record: ten successful runs per layout. [32]

Layout	Mean latency	p95
Shared-backbone distributed layout	259.4 ms	283.9 ms
Single Provider	186.8 ms	206.3 ms

This motivates measuring preparation and transfer costs. Their causal contribution was not isolated; model/export identity and raw runs require verification.

Historical record only; not evidence for the newer DI design or the active evaluation.

Backup: Provider-Transition Discovery Configurations

Provider-transition configuration	Completed
NDNSF, no pre-registered Provider addresses	357 / 357
Static-list NSC/gRPC configurations	0 / 357
Pre-registered baseline variants	357 / 357

Completion depends on how candidates are configured in these runs. Resolver or discovery integration can update baseline candidate lists. [27,28]

Backup: Historical Invocation under Packet Loss

Historical per-link loss	NDNSF completed requests
1%	599 / 600
3%	598 / 600
10% (archived run)	539 / 600

Archived NDNSF results from a separate campaign. The 10% entry is one recorded run, not an aggregate estimate; these values are not a matched baseline comparison. [28]

Per-link packet loss is not the probability of service failure: retrieval retries and the request deadline also affect completion.

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